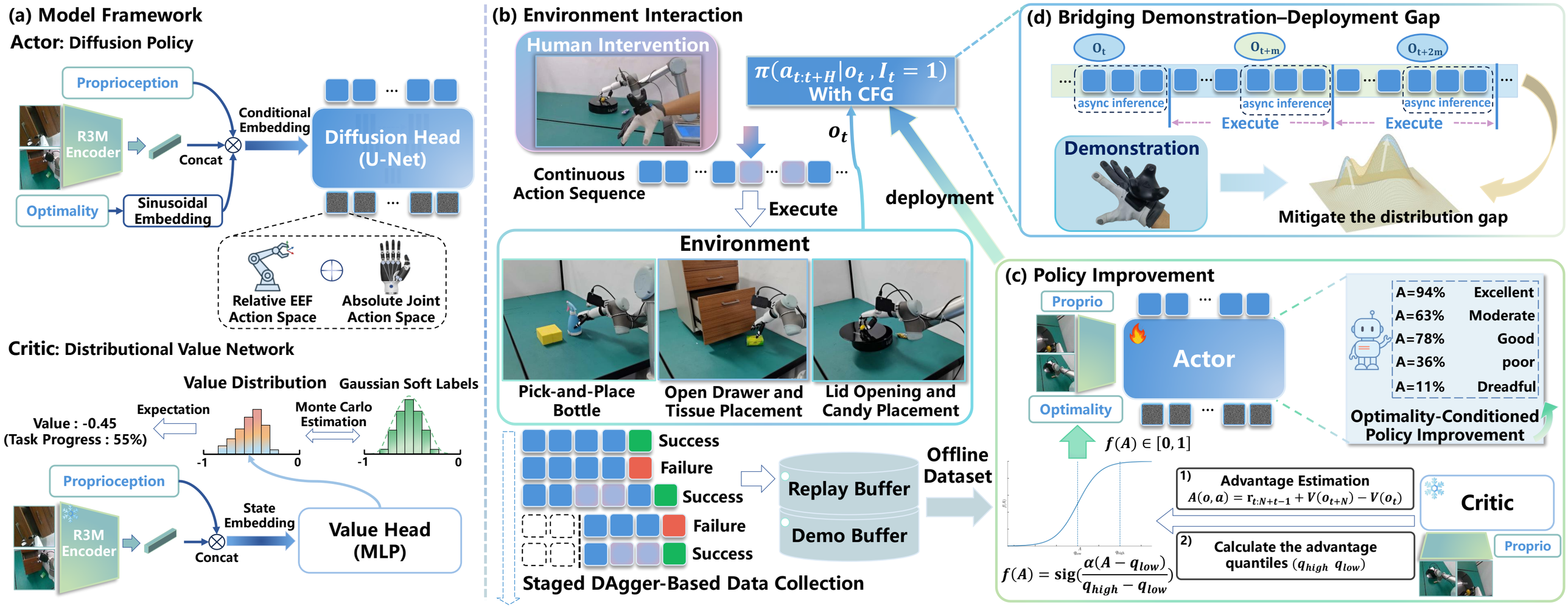
(a) Model Framework
Actor: Diffusion Policy
Proprioception
R3M Encoder
Concat
Conditional Embedding
Diffusion Head (U-Net)
Optimality
Sinusoidal Embedding
Relative EEF Action Space
Absolute Joint Action Space
Critic: Distributional Value Network
Value Distribution
Gaussian Soft Labels
Expectation
Monte Carlo Estimation
Value : -0.45
(Task Progress : 55%)
-1
0
Proprioception
R3M Encoder
Concat
State Embedding
Value Head (MLP)
(b) Environment Interaction
Human Intervention
$\pi(a_{t:t+H}|o_t, I_t = 1)$
With CFG
o_t
Continuous Action Sequence
Execute
deployment
Environment
Pick-and-Place Bottle
Open Drawer and Tissue Placement
Lid Opening and Candy Placement
Success
Failure
Success
Failure
Success
Replay Buffer
Demo Buffer
Offline Dataset
Staged DAgger-Based Data Collection
(d) Bridging Demonstration–Deployment Gap
o_t
o_{t+m}
o_{t+2m}
async inference
async inference
async inference
Execute
Execute
Demonstration
Mitigate the distribution gap
(c) Policy Improvement
Proprio
Actor
Optimality
$f(A) \in [0, 1]$
A=94% Excellent
A=63% Moderate
A=78% Good
A=36% poor
A=11% Dreadful
Optimality-Conditioned Policy Improvement
1) Advantage Estimation
$A(o, a) = r_{t:N+t-1} + V(o_{t+N}) - V(o_t)$
2) Calculate the advantage quantiles (q_{high} q_{low})
Critic
Proprio
$f(A) = \text{sig}(\frac{\alpha(A - q_{low})}{q_{high} - q_{low}})$